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National University of Computer & Emerging Sciences
FAST-Karachi Campus
CS4051- Information Retrieval
Quiz#1

Dated: February 09, 2026 Marks: 20
Time: 20 min.
Std-ID: _____Sol_____

**Question No. 1**

What are the two main assumptions we did while developing Boolean Retrieval Model? **[5]**

Assumption for Boolean Retrieval Model

1. All documents are machine readable and we can extract features from these documents.

2. Users have necessary knowledge about the features in the document and their queries are based on these features.

3. All features are equally important and independent of each other.

Effects on IR

1. Document parsing is quite challenging and it contains lot of noisy features.

2. Users have no prior knowledge of the document/ features, features can be correlated

**Question No.2**

Define the process of token normalization. Suggest what normal form should be used for these words(Tokens) – including the word itself as a possibility? **[5]**

| Token | Normalized Feature |
|---|---|
| - *O'Rourke* | orourke |
| - *Anti-discriminatory* | antidiscriminatory |
| - *U. S. A.* | USA |
| - *Co-Education* | coeducation |
| - *cont'd* | contd |

**Question No.3**

There are 16 relevant documents in a collection for a given query "q". The precision of the query is 0.40 and recall of the query is 0.25, Find, how many documents are in the results-set (number of documents response to the query "q" from the IR system)? [5]

we know,

precision = (relevant-retrieved) / (total-retrieved)

=> 0.4 = (relevant-retrieved) / (result-set)

=> (result-set) = ( relevant-retrieved) /0.4 ---------- eq(A)

similarly,

recall = (relevant-retrieved)/ (total-relevant)

=> 0.25 = (relevant-retrieved)/ 16

=> relevant-retrieved= 0.25 * 16 = 4

hence

eq(A) => result-set = 4 /0. 4 = 10

**Result-set has 10 documents.**

**Question No.4**

Explain the following type of queries from an Information Retrieval prospective with an example. Also suggest suitable data structures that get the required answer without false positive. [5]

**a. A bi-word query**

Consider the query "labor policy" it is an example of bi-word query. The intent of the user is to get the documents that contains both the words "labor" and "policy" adjacent as given in the query should be in the documents. Positional Index can be used to answer this type of query without false positives in the result-set hence no post processing required.

**b. A proximity query**

Consider the query "labor policy /k" it is proximity query. The intent of the user is to get the documents that contains both the words "labor" and "policy" within k words apart in the documents. Positional Index can be used to answer this type of query without false positives in the result-set hence no post processing required.

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**Question No. 1**

What is an Extended Boolean Model? What is the main issue of Boolean Model it tries to addressed? **[5]**

An Extended Boolean Model is a family of IR retrieval models that keeps the Boolean query style (AND/OR/NOT) but adds graded matching so documents can be partially relevant instead of only true/false. A common way to do this is to treat (i) each query term match as a weight (e.g., term frequency or normalized term weight), and (ii) combine them with a soft AND / soft OR using a parameter p that controls strictness. Extended Boolean Model addresses this by enabling partial matching + ranking while still letting users write Boolean-like queries.

**Question No.2**

Outline three differences between the two columns terms? **[5]**

| Stemming | Lemmatization |
|---|---|
| - It is a heuristic based process and different algorithms exists for any language.<br>- most of the time transform terms into non-human readable form<br>- It is extremely fast | - Use dictionary to map the term into a lexeme<br>- Human readable form.<br>- It is slow |

**Question No.3**

Suppose a query has a total of 6 relevant documents in the collection. System A and System B have each retrieved 8 documents, and the relevance status of the ranked lists is shown below:

System A: [- + - + + - - -]

System B: [+ + - - - - - +]

where the leftmost entry corresponds to the highest ranked document, and the rightmost entry corresponds to the lowest ranked document. A "+" indicates a relevant document and a "-" corresponds to a non-relevant one. For example, the top ranked document retrieved by System A is non-relevant, whereas the top ranked document retrieved by B is relevant. **[5]**

Answer the following queries:

| | |
|---|---|
| **Precision of System A** | 3/8 |
| **Precision of System B** | 3/8 |
| **Recall of System A** | 3/6 |
| **Recall of System B** | 3/6 |

Both A and B performance is same. As the Precision/Recall values do not consider the position of the relevant document in the result set.

**Question No.4**

Formulate the Information Retrieval problem mathematically, explain each terms of the formulation. **[5]**

**Mathematical formulation**

D= { $d_1$,$d_2$,…..$d_n$} contains n documents, where **n** can be very large number. Each $\mathbf{d_i}$ contains features like words/sequences / terms.

Q= { $q_1$,$q_2$,…$q_m$}

$\mathbf{R_{qi}}$ is a subset of D, relevant to the query $\mathbf{q_i}$**.** where R represents relevant score against a given query **qi** for any **dj** in D.